Exercise Sheet 24

Each exercise states the data it uses. You may use the lecture fact that $\varphi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a continuous surjective homomorphism and $d\varphi$ preserves brackets. The Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 24.1. For $x \in \mathbb{R}^4$ let $X(x) = x_0 I + x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3$. For $A \in \mathrm{SL}(2,\mathbb{C})$ define $\varphi(A) \in \mathrm{GL}(4,\mathbb{R})$ by $X(\varphi(A)x) = AX(x)A^*$. For $Y \in \mathfrak{sl}(2,\mathbb{C})$ put $d\varphi(Y) = \frac{d}{dt}\varphi(\exp tY)\big|_{t=0}$. In the basis e_0, e_1, e_2, e_3 let

$$J_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

(a) Show that $X(d\varphi(Y)x) = YX(x) + X(x)Y^*$ for $Y \in \mathfrak{sl}(2,\mathbb{C})$.

(b) Show that $d\varphi(-\frac{i}{2}\sigma_1) = J_1$ and $d\varphi(\frac{1}{2}\sigma_1) = K_1$.

(c) Compute $[K_1, K_2]$ by matrix multiplication and show that it equals $-J_3$. Compare with the bracket $[\frac{1}{2}\sigma_1, \frac{1}{2}\sigma_2]$, given that $d\varphi(-\frac{i}{2}\sigma_3) = J_3$ and $d\varphi(\frac{1}{2}\sigma_2) = K_2$.

Exercise 24.2. For $x \in \mathbb{R}^4$ let

$$X(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix},$$

and for $A \in \mathrm{SL}(2,\mathbb{C})$ define $\varphi(A)$ by $X(\varphi(A)x) = AX(x)A^*$. Let $\eta = \mathrm{diag}(1,-1,-1,-1)$ and $\langle x, y\rangle = x^T\eta y$. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(a) Compute $\Lambda = \varphi(A)$ as a 4×4 matrix.

(b) Check directly that $\Lambda^T\eta\Lambda = \eta$ and $\Lambda_{00} \geq 1$.

(c) A nonzero vector $v \in \mathbb{R}^4$ is *light-like* if $\langle v, v\rangle = 0$. Show that Λ fixes the light-like vector $e_0 + e_3$ and the vector e_2.

(d) Let $N = \Lambda - I$. Show that $N^2 \neq 0$ and $N^3 = 0$. Deduce that Λ is not diagonalisable over $\mathbb{C}$. Conclude that Λ is not conjugate in $\mathrm{GL}(4,\mathbb{C})$ to any rotation $R_3(\theta)$ of the (x_1, x_2)-plane, or to any boost $B_3(t)$, which fixes x_1, x_2 and sends $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$.

Exercise 24.3. Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Let $\rho_L(A) = A$ and $\rho_R(A) = \bar{A}$ be the two representations of $\mathrm{SL}(2,\mathbb{C})$ on $\mathbb{C}^2$.

(a) Show that $\varepsilon A\varepsilon^{-1} = (A^T)^{-1}$ for every $A \in \mathrm{SL}(2,\mathbb{C})$. Deduce that the form $\beta(\xi,\zeta) = \xi_1\zeta_2 - \xi_2\zeta_1$ satisfies $\beta(A\xi, A\zeta) = \beta(\xi,\zeta)$.

(b) Show that ρ_R is equivalent to the representation $A \mapsto (A^*)^{-1}$.

(c) Show that $\bar{U} = \varepsilon U \varepsilon^{-1}$ for $U \in \mathrm{SU}(2)$. Deduce that ρ_L and ρ_R restrict to equivalent representations of $\mathrm{SU}(2)$.

(d) Show that ρ_L and ρ_R are not equivalent as representations of $\mathrm{SL}(2,\mathbb{C})$.

Exercise 24.4. Let $W = \mathbb{C}^2 \otimes \mathbb{C}^2$ carry the representation $(\frac{1}{2},\frac{1}{2})$, namely $A \mapsto A \otimes \bar{A}$ of $\mathrm{SL}(2,\mathbb{C})$. Let $\Lambda^2 W$ carry the induced action $u \wedge v \mapsto (A\otimes\bar{A})u \wedge (A\otimes\bar{A})v$. Use the following facts from the lecture.

The complexified differential of (j_L, j_R) is the module $V_{2j_L} \boxtimes V_{2j_R}$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, where (Y_1, Y_2) acts by $dS^{2j_L}(Y_1) \otimes 1 + 1 \otimes dS^{2j_R}(Y_2)$. Here $V_m = \mathrm{Sym}^m \mathbb{C}^2$ and S^m is the induced symmetric-power representation.

A linear map intertwines two finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ if and only if it intertwines their complexified differentials. Let $H = \mathrm{diag}(1,-1)$, $h_L = (H,0)$ and $h_R = (0,H)$. A *weight* is a pair of simultaneous eigenvalues of (h_L, h_R).

You may also use the lecture's restriction and descent rules: (j_L, j_R) restricts to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, \dots, j_L + j_R$ in integer steps, and descends precisely when $2j_L + 2j_R$ is even.

The four-vector identification uses $\Psi(\xi \otimes \zeta) = \xi\zeta^T$ and the complex-linear extension $X_{\mathbb{C}}$ of the map X in Exercise 24.1.

(a) List the weights of W on the basis $e_i \otimes e_j$.

(b) List the weights of $\Lambda^2 W$, with multiplicities.

(c) Deduce that $\Lambda^2 W \cong (1,0) \oplus (0,1)$ as representations of $\mathrm{SL}(2,\mathbb{C})$. You may use the Lie algebra background: finite-dimensional modules of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ are completely reducible, and $V_a \boxtimes V_b$ has highest weight (a,b).

(d) The lecture identifies W with $\mathbb{C}^4$, carrying the complexified four-vector representation. Deduce the dimension and the $\mathrm{SU}(2)$-content of $\Lambda^2\mathbb{C}^4$. Decide whether it descends to $\mathrm{SO}^+(1,3)$.

Exercise 24.5. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \dots\}$ let $D^{(j_L,j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A})$ on $V_{2j_L} \otimes V_{2j_R}$. Here $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$ and $S^m(A)$ is the restriction of $A^{\otimes m}$. Use the lecture's rules: the representation descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L + 2j_R$ is even, and its restriction to $\mathrm{SU}(2)$ is $\bigoplus V_{2j}$ for $j = |j_L - j_R|, \dots, j_L + j_R$ in integer steps.

(a) List all pairs (j_L, j_R) with $\dim D^{(j_L,j_R)} \le 4$. For each, give the restriction to $\mathrm{SU}(2)$ and decide whether it descends.

(b) Show that all spins j in the restriction satisfy $2j \equiv 2j_L + 2j_R \pmod 2$. Check this against the action of $-I \in \mathrm{SU}(2)$.

(c) Show that $D^{(j_L,j_R)}$ has a nonzero $\mathrm{SU}(2)$-fixed vector if and only if $j_L = j_R$. Show that the fixed space is then one-dimensional.

(d) The lecture realises $(\frac{1}{2},\frac{1}{2})$ on $M_2(\mathbb{C})$ by $Z \mapsto AZA^*$. Find its $\mathrm{SU}(2)$-fixed line directly.

Exercise 24.6. Let V be a finite-dimensional complex space with an inner product $\langle \cdot, \cdot \rangle$, antilinear in the first slot. Let Π be a continuous representation of $\mathrm{SL}(2,\mathbb{C})$ on V with every $\Pi(A)$ unitary. You may use the following facts. Π is smooth (Lecture 3), and $\Pi(\exp Y) = \exp d\Pi(Y)$ for $Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ (Lecture 2).

Every irreducible finite-dimensional continuous representation is equivalent to some $D^{(j_L,j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A})$. Moreover, $dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y})$ with $m = 2j_L$ and $n = 2j_R$. The element $H = \mathrm{diag}(1,-1)$ acts on V_m with eigenvalues $m, m-2, \ldots, -m$.

(a) Show that $d\Pi(Y)$ is skew-Hermitian for each $Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$, so its eigenvalues are purely imaginary.

(b) Compute the eigenvalues of $dD(\frac{1}{2}\sigma_3)$ on $D^{(j_L,j_R)}$. Deduce that $D^{(j_L,j_R)}$ admits an invariant inner product only if $(j_L, j_R) = (0,0)$.

(c) Deduce that Π is trivial. Deduce the same for finite-dimensional continuous unitary representations of $\mathrm{SO}^+(1,3)$.

(d) Compute the eigenvalues of $dD(-\frac{i}{2}\sigma_3)$. Explain why the argument of (b) gives no obstruction for $\mathrm{SU}(2)$.